EP curs 2 – Common mistakes and metrics

Common mistakes

1. No goals
2. Biased goals
3. Unsystematic approach
4. Analysis without understanding the problem
5. Incorrect performance metrics
6. Unrepresentative workload
7. Wrong evaluation technique
8. Overlook important parameters
9. Ignore significant factors
10. Inappropriate experimental design
11. Inappropriate level of detai
12. No analysis
13. Erroneous analysis
14. No sensitivity analysis
15. Ignoring errors in input
16. Improper treatment of outliers
17. Assuming no change in the future
18. Ignoring variability
19. Too complex analysis
20. Improper presentation of results
21. Ignoring social aspects
22. Omitting assumptions and limitation

A systematic approach to pert evaluation

1. State goals and define the system
2. List services and outcomes
3. Select metrics
4. List parameters
5. Select factors to study
6. Select evaluation technique
7. Select workload
8. Design experiments
9. Analyze and interpret data
10. Present results

    Repeat

Example

1. System definition
   The network connecting two hosts.
2. Services
   Instant messaging application (including file transfers).
3. Metrics
   - No errors or failures.
   - Rate (throughput), time (latency, rtt), resource per service.

   Leads to these metrics:
   i. Time per message or file
   ii. Maximum transfer rate per unit of time
   iii. Local and remote CPU time per transmission
4. System parameters
   - Speed of local CPU, remote CPU, network
   - Operating system
   - Network reliability
5. Workload parameters
   - Time between successive message/file transmissions
   - Sizes of messages/files
   - Type of channel
   - Other loads on the local and remote CPUs and the network
6. Factors
   - Type of channel
   - Size of network
   - Size of message/file transmitted
   - Keep fixed: type of CPU, OS, no other loads on hosts and network
7. Evaluation technique
   - Implement prototype => measurements
   - Use analytical modeling for validation
8. Workload
   - Synthetic program generating the specified types of channel requests
   - Null channel requests => resources used in monitoring and logging
9. Experimental design
   - A full factorial experiment design e.g. 2*3*3
10. Data Analysis
11. Data Presentation

Summary

- Analysis techniques, metrics, workload - depend on the goal
- Metrics are based on the services provided by the system
- System and workload parameters determine the right set of experiments

Metrics

Selecting evaluating techniques

Selecting metrics

Techniques

- Mathematical Modeling
- Simulation
- Measurement

Mathematical Modeling

- Skill set required
  - Math
- Abstraction of system
  - High-level
- Assumptions
  - Many
- Results
  - Asymptotic results, low accuracy
- Outliers
  - Not captured

Simulation

- Represents an almost "real-world" view of the system
- System can be modeled as a set of interacting subsystems and objects
- Various levels of abstraction possible
- Fine-grained performance metrics can be measured
- Coding errors
- Does not capture real-world environment

Measurement

- Testbed
- Experimental results

Selecting and evaluation technique

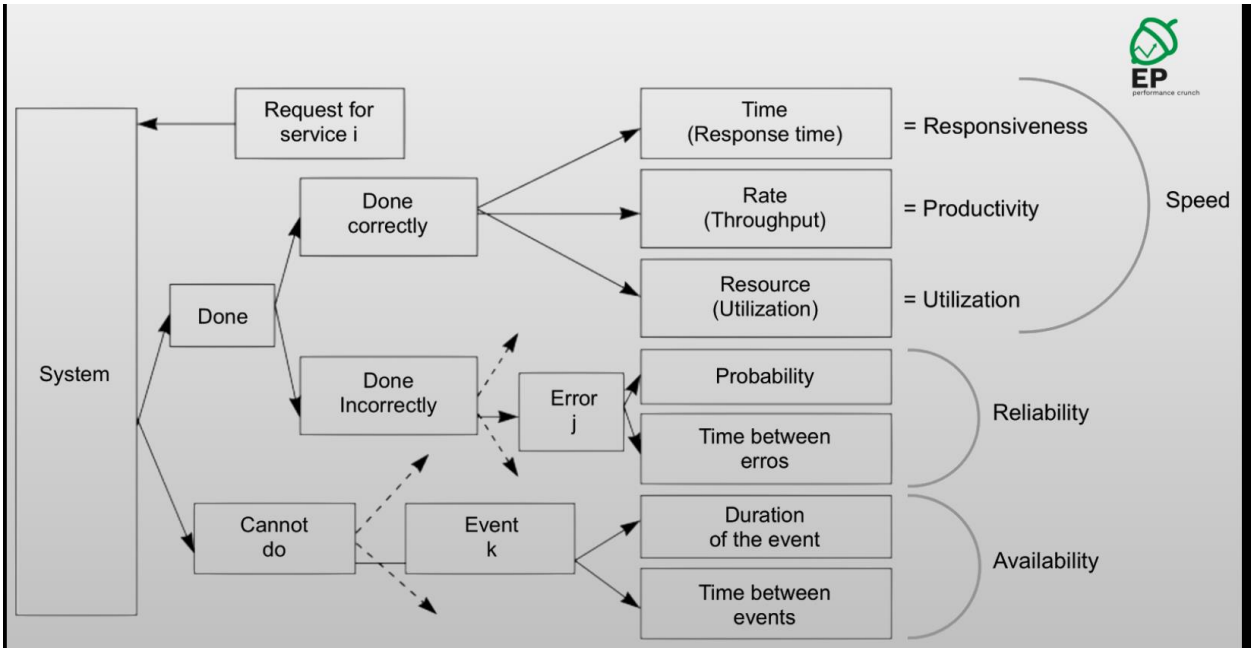
Criterion	Analytical Model	Simulation	Measuring
1. Stage	Any	Any	Prototype
2. Time required	Small	Medium	Varies
3. Tools	Analysts	Computer languages	Instrumentation
4. Accuracy	Low	Moderate	Varies
5. Trade-off evaluation	Easy	Moderate	Difficult
6. Cost	Low	Medium	High
7. Saleability	Low	Medium	High

Rules of validation

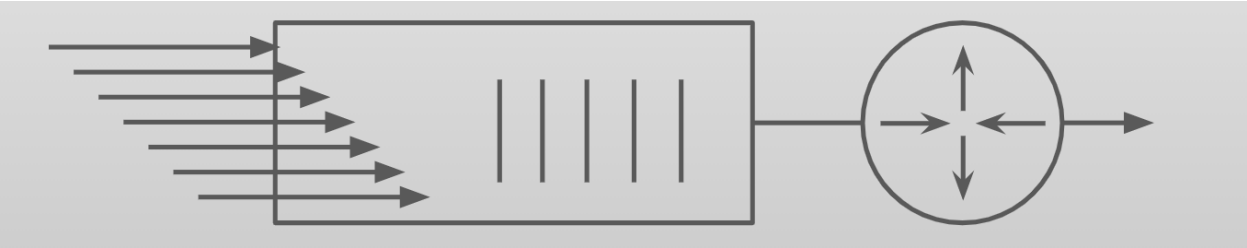
- Do not trust the results of **a simulation model** until validated with analytical modeling or measurements
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- Do not trust the results of **a measurement** until validated with simulation or analytical modeling

Performance metrics

- List services and outcomes

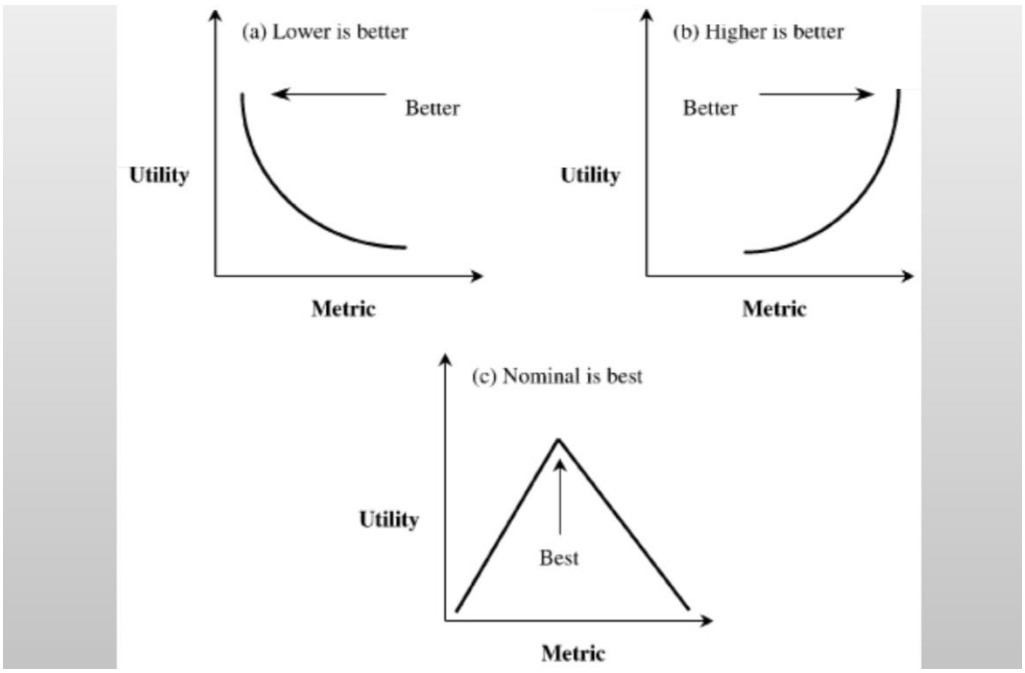


Global vs. individual



Common metrics

- Response time
  - Interval between a user's request and the system response
- Turnaround time
  - For batch stream
  - Time between the submission of a batch job and the completion of its output
- Throughput
  - The rate (requests per unit of time) at which the requests can be serviced by the system
- Nominal capacity
  - The maximum achievable throughput under ideal workload conditions
  - Bandwidth
- Usable capacity
  - The maximum throughput achievable without exceeding a prespecified response time limit (Often the response time at maximum throughput is too high to be acceptable.)
- Efficiency
  - Usable capacity / nominal capacity
- Utilization
  - The fraction of time the resource is busy servicing requests
- Reliability
  - The probability of errors or the mean time between errors
- Availability
  - The fraction of the time the system is available to service users' requests
  - Mean Time To Failure (MTTF) - mean uptime
- Cost/performance ratio
  - Metric for comparing systems
  - E.g. two transaction processing systems may be compared in terms of dollars per TPS



Requirements

- "The system should be both processing and memory efficient. It should not create excessive overhead."
- "There should be an extremely low probability that the network will duplicate a packet, deliver a packet to the wrong destination, or change the data in a packet."

- Unacceptable
  - Non-Specific
  - Non-Measurable
  - Non-Acceptable
  - Non-Realizable
  - Non-Thorough